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Epidemiological Note

Significance of impaired glucose tolerance in an Asian Indian population: a follow-up study

Ambady Ramachandran, Chamukuttan Snehalatha, Ramavat Amara Singh Naik, Viswanathan Mohan, Ramachandran Shobana and Moopil Viswanathan

Diabetes Research Centre, 5, Main Road, Royapuram, Madras 600013, India

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Summary

A prospective study was undertaken in 107 Indians with impaired glucose tolerance (IGT) for a period ranging from 2 to 10 years. On follow-up, 32% still had an impaired glucose tolerance, 32% reverted to normal glucose tolerance and 36% developed diabetes. Careful dietary adherence and weight reduction were found to favour normalisation of glucose tolerance. Poor dietary adherence, persistent obesity and weight gain were found to precipitate diabetes. The study stresses the need for intensive diet therapy in individuals with IGT.

Insulin responses were heterogenous in IGT and non-predictive of the follow-up changes in glucose tolerance.

Introduction

Individuals with impaired glucose tolerance (IGT) are considered to be intermediate between normal and diabetic individuals. The reasons for this special classification by the National Diabetes Data Group (NDDG) [1] and the WHO Expert Committee [2] are that individuals with IGT show slow progression to diabetes, low frequency of diabetic symptoms and microvascular complications and frequent regression to normal glucose tolerance. There have been few longitudinal studies on individuals with IGT [3–5] and even these studies have been done in European and Japanese populations.

There are striking differences in the pattern of non-insulin-dependent diabetes mellitus (NIDDM) seen in India compared to that in Europeans [6,7]. These differences include reversal of sex ratio, younger age at onset and lower prevalence of obesity. It is likely that the significance of IGT in one population could be different from that in another. Hence, we analysed the outcome in a group of individuals having IGT according to the NDDG criteria. [1].

Address for correspondence: Dr. A. Ramachandran, M.D., Asst. Director, Diabetes Research Centre, 5, Main Road, Royapuram, Madras 600013, India.

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Material and methods

All subjects who showed IGT during their initial visit to the Centre and who were followed up for periods ranging from 2 to 10 years were chosen for the study. Out of a total 107 there were 74 males and 33 females and their age ranged from 35 to 72 years. There were 35 obese and 72 non-obese individuals.

All of them underwent thorough clinical and biochemical examinations including oral GTT, liver function and kidney function tests, ECG and chest X-ray. Oral GTT was performed with 75 g glucose load. Blood samples were collected fasting and at half-hour intervals after glucose intake, for 2 h. A detailed case history was recorded with special emphasis on family history of diabetes mellitus. The body mass index (BMI) was calculated using the formula (weight in kg)/(square of the height in metres). A BMI of $\geq$ 27 was considered to indicate obesity in men and $\geq$25 in women. A change in weight by a minimum of 2 kg during the follow-up period was considered to indicate weight gain or loss.

All individuals were advised a calorie-restricted high carbohydrate high fibre diet, the details of which have been previously reported [8]. Those who were obese were advised weight reduction. The total calorie content in the diet varied from 1400 to 2000 kcal/day depending on the individual needs and the diet consisted of 60–70% carbohydrate, 15% protein and 15% fat. At each subsequent visit, the patient's diet was analysed in detail by a dietitian, using a scoring system developed by us [9] and the diet adherence was assessed. Plasma glucose was estimated by the *O*-toluidine method using a venous sample.

Patients were reviewed periodically with plasma glucose estimations. The classification of the individuals was based on the values of final GTT on follow-up. Those who still had IGT were classified as Group I (IGT), those with normal values as Group II (normal) and those with diabetes as Group III (diabetic).

In 31 individuals immunoreactive insulin (IRI) was measured during the initial and final GTTs. The serum samples were preserved at −20°C for IRI estimation by the procedure of Herbert et al. [11].

The area under the insulin curve (ΣIRI) was calculated by adding the IRI values at 5 time intervals during GTT. The sum of the glucose values (ΣG) was also obtained by adding the 5 glucose values during GTT. The ratio of ΣIRI/ΣG was calculated for each person and indicated as the I/G ratio. The values were compared with respective control values for obese and non-obese individuals.

Statistical comparisons were made using the *t*-test, chi-square test and Fisher's exact probability test, where relevant.

Results

The results were analysed in relation to the following parameters: (a) positive family history of diabetes; (b) obesity; (c) change in body weight during follow-up; and (d) diet adherence; (e) a combination of the above factors.

Plasma glucose

The plasma glucose values of the 3 groups are shown in Fig. 1. On follow-up, 34 (32%) individuals still had IGT (Group I), 34 (32%) reverted to normal (Group II), while the remaining 39 (36%) became diabetic (Group III). The mean period of follow-up for each group was 4.5 ± 3.5, 3.3 ± 2.0 and 5.1 ± 3.5 years, respectively.

Among the 39 individuals who became diabetic, 16 developed diabetes within 3 years, 12 within 4–5 years and 11 after 5 years of follow-up. At the start of the study the mean age of the individuals in the 3 groups was 48.6 ± 5.9, 47.7 ± 5.8 and 48.7 ± 6.2 years, respectively (mean ± SD).

Analysis of the different parameters influencing diabetogenesis was made in the 3 groups and Table 1 shows the results, with the statistical comparisons.

Family history of diabetes (FH)

Among the 107 individuals, FH was positive in 73 (68%) and among these 73, 19 individuals developed diabetes. Comparing the groups with and

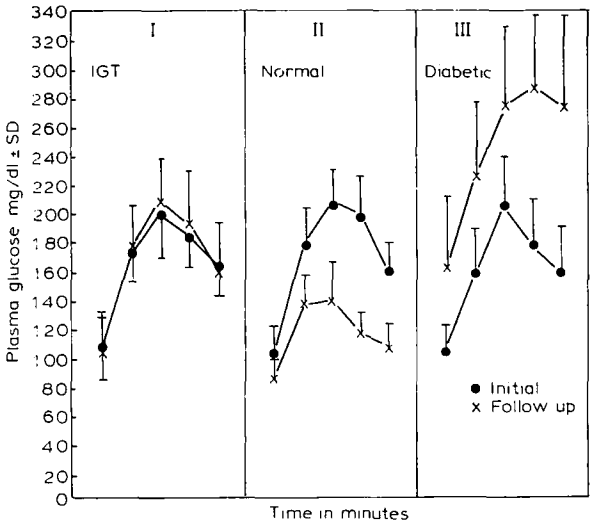

Fig. 1. Plasma glucose values in the 3 groups of IGT individuals.

without positive family history of diabetes, there was no statistical difference in the percentage becoming diabetic in either group (chi-square test).

Obesity

Inititally, obesity was present in 35 (33%) of the study group and 46% of these became diabetic on follow-up. Comparing the obese and non-obese groups, diabetogenesis was found to be significantly higher in the obese group (Fisher's test).

During the follow-up, 8 obese persons lost weight (5–8 kg) and 2 of them still had IGT, 5 became normal and only one became diabetic.

Gain in body weight

In the diabetic group, 23% had gained weight during the period of follow-up. In the normal group, 15% gained weight but remained within the ideal body weight range (Table 2).

Gain in body weight was seen in 15 (31%) of the non-obese individuals. Of these, 5 still had IGT, 5 deteriorated to diabetes and 5 became normal. Gain in body weight in overweight subjects favoured the development of diabetes.

TABLE I

PARAMETERS INFLUENCING DIABETOGENESIS OR REMISSION IN IGT

Parameters	Total	Positive		Total	Negative		Test used and significance	Remarks
		Normal	Diabetic		Normal	Diabetic		
1. Family history	73	20	19	34	14	20	χ^2 — non-significant	Family history has no direct influence on diabetogenesis
2. Obesity	35	5	16	72	29	23	Fisher's exact probability test — significant at 5% level	Persistent obesity precipitates diabetogenesis
3. Weight gain	25	5	9	82	29	30	Fisher's exact probability test — significant at 5% level	Weight gain favours diabetogenesis
4. Weight loss	42	17	15	65	17	24	Fisher's exact probability test — significant at 5% level	Favours remission
5. Diet adherence	52	31	10	55	3	29	χ^2 — highly significant ($P < 0.001$)	Lack of diet adherence precipitates diabetes. Diet adherence favours remission

TABLE 2

INFLUENCE OF CHANGE IN BODY WEIGHT ON IGT

Numbers in brackets are percentages. All values are mean ± SD.

Follow-up GTT	Initial obesity (No.)	Weight gain (No.)	Initial BMI[a]	Follow-up BMI[a]	Weight loss (No.)	Initial BMI[a]	Follow-up BMI[a]
IGT (n = 34)	14 (39)	11 (32)	27.6 ± 2.1	29.2 ± 2.3	10 (30)	27.2 ± 2.2	25.2 ± 2.0
Normal (n = 34)	5 (15)	5 (15)	20.3 ± 2.4	22.9 ± 2.2[a]	17 (50)	25.2 ± 2.5	22.8 ± 2.1[a]
Diabetic (n = 39)	16 (41)	9 (23)	25.2 ± 2.2	27.8 ± 2.2	15 (39)	29.3 ± 3.9	26.8 ± 2.0

[a] These changes were within the non-obese range.

Weight reduction

Weight reduction favoured reversal of IGT to normal. In the normal group, 50% had weight reduction. Loss of even 2–3 kg body weight brought about a significant improvement in the blood glucose curve. In the diabetic group 39% showed weight loss and this was after the development of the disorder and also related to its severity (Σglucose 2180 ± 40 mg against 1140 ± 48 mg in the others).

Diet adherence

Adherence to the diet prescribed was found to be an important factor influencing remission of IGT. 91% of the individuals in the normal group showed good diet adherence and only 26% in the diabetic group had adhered to the diet prescriptions.

Poor diet adherence was recorded in 55 (51%) individuals. Among these, 29 (52%) developed diabetes. Statistical comparison of those who followed the diet prescription with those who did not, showed that diabetogenesis was strongly associated with poor diet adherence.

The combined effect of multiple factors on diabetogenesis was observed in several individuals, as shown in Table 3.

TABLE 3

COMBINED EFFECT OF DIABETOGENIC FACTORS

Obesity	+ FH[a]	Poor diet adherence	Number of individuals
+	+	–	8
+	–	+	14
+	+	+	9
–	+	+	7

[a] FH = Family history.

Insulin responses

Insulin responses to oral glucose load were heterogenous in persons with IGT.

The initial I/G ratios were normal (19%), low (42%) or high (39%), as shown in Table 4. In the normal group, with the decrease in blood glucose values, the percentage of high insulin responses decreased. In low and normal insulin responders, I/G ratios remained unchanged.

The initial and follow-up I/G ratios in the 3 groups of patients are shown in Fig. 2. In the IGT

TABLE 4

FOLLOW-UP GTT PATTERN IN THE 3 TYPES OF INSULIN RESPONSES

Numbers in brackets are percentages.

Initial I/G ratios		Diabetic on follow-up
Normal ± 1 SD		Non-obese 0.4 –0.5 Obese 1.08–2.0
Low	13 (42)	4 (31)
Normal	6 (19)	1 (17)
High	12 (39)	3 (25)

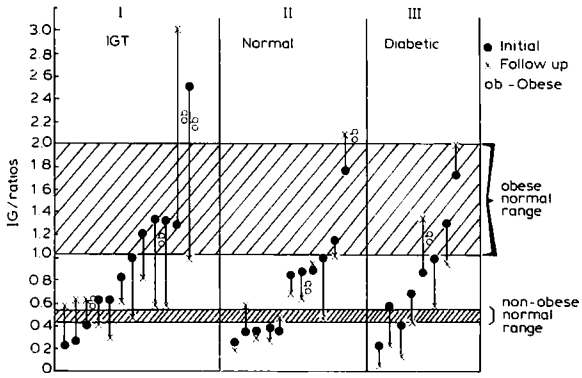

Fig. 2. Initial and follow-up I/G ratios in the 3 groups.

group and the diabetic group there was a tendency for the insulin values to decrease on follow-up in spite of the blood glucose values remaining high. In 3 patients, there was a significant increase in I/G ratio and this was associated with weight gain.

Discussion

This study shows that Indians with IGT run a high risk of developing diabetes. 36% of persons with IGT developed diabetes in a mean period of 4 years and 34% still had IGT. However, GTT had reverted to normal in 34% of the group. Careful and continued dietary regulations as well as correction of body weight seem to be the 2 most important factors in preventing deterioration of carbohydrate tolerance. This underscores the need for instituting active dietetic measures in persons with IGT. Ohnede et al. [5] from Japan also noted that 35% of their study group with initial 2 h post-glucose values between 135 and 195 mg/dl developed diabetes in 5 years. The need for strict dietetic therapy was stressed by the authors.

The diabetic diet used in our centre is a high carbohydrate high fibre diet. Major emphasis is laid on the restriction of total calories and avoidance of refined sugars [8,9]. Our long-term studies have proved that the diet is effective in improving carbohydrate tolerance [12,13]. The detrimental effects of excessive calorie intake have also been well brought out by the studies of West [14]. Sartor et al. have shown that oral antidiabetic drugs are also useful in a selected group with IGT [3].

Persistent obesity as well as weight gain were found to enhance the risk of developing diabetes in persons with IGT. In this study, we found that gain in weight in overweight persons favoured diabetogenesis. Keen et al. in the Bedford Survey [4] had noticed that obesity doubled the risk of deterioration of IGT to diabetes. Gain in weight in non-obese subjects who remained within the IBW range did not influence progression to diabetes.

Decrease in body weight, even to a small degree, helped in improving glucose tolerance in both obese and non-obese groups. This phenomenon has been observed earlier in NIDDM patients [4]. As the study group comprised of individuals with similar baseline values for plasma glucose, the impact of the initial degree of hyperglycemia could not be studied. The mean age of the individuals in the 3 different groups was comparable and hence the effect of age on further deterioration of glucose tolerance was not considered in this study.

Family history of diabetes does not appear to influence the deterioration to overt diabetic state. However, it has to be mentioned that recording of negative family history by questionnaire method has inherent problems. All efforts have been made to trace the family history back to 3 generations, in all patients.

Insulin secretion showed a heterogenous pattern in IGT. The majority of them (81%) showed an abnormal insulin/glucose ratio, either due to an inability to secrete appropriate amounts of insulin for the elevated blood glucose (low response in 42%) or due to an exaggerated insulin output (high response in 39%).

As in non-insulin-dependent diabetes, a low response may indicate a B-cell defect whereas a high response may be due to insulin resistance. These abnormalites, however, do not predict the course of IGT during the follow-up period as they are modified by therapeutic and environmental factors.

In the group which showed normalisation of blood glucose, the normal I/G ratios showed no variations during the follow-up, indicating that the B-cells were able to respond appropriately to the

glucose levels in blood. The improvement in the glucose values in spite of low I/G ratios probably indicates enhanced insulin sensitivity.

All the exaggerated I/G ratios seen initially showed a reduction, irrespective of the plasma glucose on follow-up. The fall in I/G ratio with normalisation of plasma glucose indicates a decrease in insulin resistance. On the contrary, this would probably indicate a deterioration in the B-cell secretory capacity if the plasma glucose remained high.

In summary, the study shows that IGT in Indians may be an early phase of diabetes and progression to diabetes occurs if proper therapeutic measures are not instituted. On the contrary, the biochemical abnormalities can be reverted to normal by instituting these measures.

References

1 National Diabetes Data Group. Classification of diabetes mellitus and other categories of glucose intolerance, *Diabetes*, 28: 1039–1057, 1979.

2 WHO Expert Committee on Diabetes Mellitus, *WHO Tech. Rep. Second Rep. Ser.* 646, 1980, pp. 7–14.

3 Sartor, G., Schersten, B., Carlstorm, S., Melander, A., Norden, A. and Person, G., Ten year folow-up of subjects with impaired glucose tolerance. Prevention of diabetes by tolbutamide and diet regulation, *Diabetes*, 29: 41–49, 1980.

4 Keen, H., Jarrett, R. J. and McCartney, P., The ten year follow-up of Bedford Survey (1962–1972). Glucose tolerance and diabetes, *Diabetologia*, 22: 73–78, 1982.

5 Ohnede, A., Kobayashi, T. and Nihei, J., Evaluation of new criteria for diagnosis of diabetes mellitus based on follow-up study of border line diabetes, *Tohoku J. Exp. Med.*, 137: 437–444, 1982.

6 Mohan, V., Ramachandran, A. and Viswanathan, M., Tropical diabetes. In: K.G.M.M. Alberti and L.P. Krall (eds.), *The Diabetes Annual – 1*, Elsevier, Amsterdam, 1985, pp. 82–92.

7 West, K.M., Diabetes in the tropics. Some lessons for Western diabetology. In: S. Podolsky and M. Viswanathan (eds.), *Secondary Diabetes. The Spectrum of Diabetes Syndromes*, Raven Press, New York, 1980, pp. 249–255.

8 Viswanathan, M., Ramachandran, A., Mohan, V. and Snehalatha, C., High carbohydrate high fibre diet in diabetes, *J. Diabetic Assoc. India*, 30: 90–96, 1981.

9 Mohan, V., Ramachandran, A., Sheela Paul et al., Assessment of dietary adherence in diabetic subjects by a new scoring system, *Antiseptic*, 81: 133–136, 1984.

11 Herbert, V., Lau, K.S., Gottleib, G.W. and Bleicher, S.J., Coated charcoal immunoassay of insulin, *J. Clin. Endocrinol. Metab.*, 25: 1375–1384, 1965.

12 Viswanathan, M., Snehalatha, C., Ramachandran, A. and Mohan, V., High carbohydrate diet in diabetes. Long term experience. In: W. Waldhausl (ed.), *Proceedings of the X International Diabetes Federation Congress, Vienna*, Excerpta Medica, Amsterdam, 1979, p. 84.

13 Viswanathan, M., Mohan, V., Ramachandran, A., Snehalatha, C. and Anderson, J.W., Long-term experience with high carbohydrate high fibre diets in Indian diabetic patients, *Diabetol. Croat.*, 13: 163–174, 1984.

14 West, K.M. (ed.), *Epidemiology of Diabetes and its Vascular Lesions*, Elsevier, New York, 1978, p. 249.